

ENTRY 027 Retiring `variable_forgiveness_ratio()` — August 13, 2026 — Status: COMPLETE

Entry 027 — The Fitted Correction Comes Out, the Derived One Goes In

Generalizing Entry 025's Closed Form to Real Multi-Edge Routes, and Checking It Doesn't Cost Anything

1. What Was Done

Entry 026 found no clean residual left after subtracting the Entry 025 closed-form baseline from every recorded decay curve — the empirically fitted `variable_forgiveness_ratio()` power law was mostly re-deriving that baseline, not new chip physics. Its recommendation was to retire the fitted correction and use the derived formula directly. This entry does that.

Entry 025's closed form was derived for one edge's SAME depolarizing channel applied n times. Real routes cross several different edges with different raw errors, so the absorbing-state argument needed generalizing first: each two-qubit gate application either fires the channel (probability p_i , sending the state to maximally mixed) or doesn't (probability $1-p_i$) — independent of every other gate, and firing anywhere is absorbing. So the state survives untouched with probability $\prod_i (1-p_i)$, and expected success is $\text{floor} + (1-\text{floor}) \cdot \prod_i (1-p_i)$, where floor is the ideal outcome set's share of the full outcome space (0.5 for a 2-qubit Bell pair, 0.25 for a 3-qubit GHZ state). No fitted coefficients — only the raw calibrated error and a combinatorial floor.

Implemented in v4 (`route_cost/predict`) and the exact-dwell router (`exact_dwell_cost`): gate-error survival is now accumulated as one product across the whole circuit and converted to a probability exactly once via the floor formula, instead of being forgiveness-corrected gate-by-gate and multiplied straight into a running success value. Decoherence, single-qubit gate cost, and readout are untouched — they were never part of the forgiveness fit.

2. Re-Validation Against the Entry 022 Kyiv Reference

circuit	Aer ref	v4.1 (old, fitted)	v4.1 (new, closed-form)	Δ
bell_adjacent_77_78	96.07%	gap 0.05	gap 0.09	+0.04
bell_near_77_82	48.74%	gap 2.39	gap 2.68	+0.29
bell_mid_77_100	83.69%	gap 19.07	gap 19.11	+0.04
bell_far_0_126	69.90%	gap 45.83	gap 43.79	-2.04
ghz_local_77_78_79	88.53%	gap 0.32	gap 0.76	+0.44
MEAN		13.54	13.28	-0.26
MEDIAN		2.39	2.68	+0.29
WORST		45.83	43.79	-2.04

All gaps in percentage points. Mean and worst-case both moved slightly in the derived model's favor; median moved slightly against it. The differences are within a point either way — a wash, not a regression.

Entry 027 — Conclusion

Swapping the fitted `variable_forgiveness_ratio()` power law for the derived closed-form floor changes accuracy on the Entry 022 Kyiv reference set by less than a point in either direction on every circuit tested (mean gap $13.54 \rightarrow 13.28$, worst-case $45.83 \rightarrow 43.79$). That confirms Entry 026's diagnosis directly: the fitted ratio was buying essentially nothing beyond what the derived formula already provides for free, while carrying two chip-specific fitted coefficients, an explicit fit-range caveat, and a dependency on future recalibration for every new chip. The derived version needs none of that — it falls out of the physics of a depolarizing channel and the size of the circuit's ideal outcome set.

3. An Honest Complication, Not Fixed by This Entry

bell_mid_77_100 (19-point gap) and bell_far_0_126 (44-point gap) remain badly wrong under both the old and new models, essentially unchanged by this substitution. Both are the two longest SWAP chains in the reference set (bell_far alone crosses 25 real swap legs). This is a pre-existing, unsolved problem — not something Entry 027 caused or was expected to fix, since it only touched how gate error is aggregated, not the routing or decoherence-integration logic Entries 023/024 built. It is flagged here rather than left implicit: whatever is wrong with the emulator's long-route predictions is still wrong, and is the most promising lead for a future entry.

4. Next Step — Entry 028

Investigate the long-route gap directly: bell_mid_77_100 and bell_far_0_126 are the two worst-predicted circuits in the entire reference set, by a wide margin, and share the trait of crossing many SWAP legs. Candidates to isolate: whether decoherence integrated over 20+ sequential legs is itself miscalibrated at that length, whether Aer's own routing diverges from the exact-dwell router's choice on long paths under a different seed, or whether an effect this project hasn't modeled yet (crosstalk between simultaneously-idle spectator qubits, for instance) only becomes visible once a route is long enough.

Research Log — Index Addendum (Entry 027)

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Entry 027	Retiring variable_forgiveness_ratio() for the Closed-Form Floor	Aug 13, 2026	Complete

variable_forgiveness_ratio() and its two fitted coefficients (forgiveness_law.json) are superseded in the v4/v4.1 codepath by emulator_v4.ideal_set_floor(), a derived closed form requiring no calibration data to fit. Retained in emulator_v3_routing.py unmodified, used only as the historical v3 baseline for comparison.